

Technical Approach

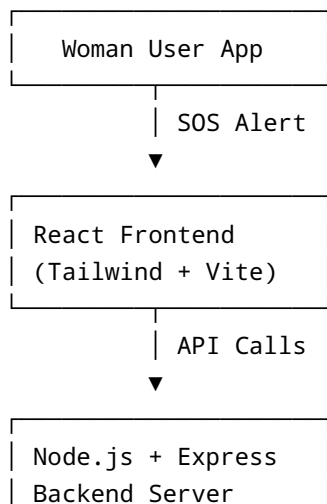
System Architecture

```
graph TD;
    A[Woman User App] --> B[React Frontend (Vite + Tailwind)];
    B --> C[Node.js + Express Backend API];
    C --> D[MongoDB Database];
    D --> E[Socket.io Realtime Server];
    E --> F[Nearby Volunteers];
```

External Integrations

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Nyckel AI → Woman Verification
ApyHub API → Aadhaar Verification
Leaflet Maps → Live Tracking & Navigation
Firebase FCM → Push Notifications
```

Architecture Diagram





Tech Stack

Technology	Purpose	Why Chosen
React.js	Frontend UI	Fast, component-based architecture
Vite	Build Tool	Faster development & hot reload
TailwindCSS	Styling	Rapid responsive UI development
Node.js	Backend Runtime	Scalable realtime handling
Express.js	API Framework	Lightweight & flexible APIs
MongoDB	Database	Flexible schema for alerts & users
Socket.io	Realtime Communication	Instant SOS delivery & live tracking
JWT Auth	Authentication	Secure role-based access
Leaflet Maps	Live Maps & Navigation	Lightweight realtime map rendering
Nyckel AI API	Woman Verification	AI-based identity validation
ApyHub API	Aadhaar Verification	Secure volunteer verification
Firebase FCM	Push Notifications	Instant emergency notifications

Data Flow / Working Pipeline

SOS Alert Workflow

1. Woman triggers SOS alert
2. Frontend sends alert to backend API

3. Backend stores alert in MongoDB
 4. Socket.io broadcasts alert to nearby active volunteers
 5. Volunteer accepts alert in realtime
 6. Woman receives volunteer profile + live location
 7. Volunteer location updates every 8 seconds
 8. Live route shown using Leaflet maps
 9. Alert marked resolved after assistance
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Realtime Rescue System

Core Realtime Features

- Instant SOS broadcasting
 - Volunteer live tracking
 - Realtime location updates
 - Socket room-based communication
 - Live navigation route rendering
 - Emergency acceptance acknowledgements
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Security & Verification

Safety Measures

- JWT secured authentication
- AI-based woman verification using webcam selfie
- Aadhaar verification for volunteers
- Role-based protected routes
- Verified volunteer profiles
- Secure realtime socket communication